CNS Lab

Assignment-1

Problem 1: Given the 16-bit Hexadecimal number: **`0xA3C5`**

1. Convert `0xA3C5` to 16-bit binary.
2. Left shift (LF) by 4 bits. (Give Binary and Hex)
3. Right shift (RF) by 4 bits. (Give Binary and Hex)
4. Left circular shift by 4 bits. (Explain wrapped bits)
5. Right circular shift by 4 bits. (Explain wrapped bits)

Problem 2: Given the 16-bit Binary number: **`1000 0000 0000 0011`**

1. Left shift (LF) by 2 bits. (State what happens to the highest bit and what fills the bottom)
2. Right shift (RF) by 3 bits. (Give Binary)
3. Left circular shift by 2 bits. (Show steps)
4. Right circular shift by 3 bits. (Show steps)

Problem 3: Given the 16-bit Hexadecimal number: **`0x7F03`**

1. Convert `0x7F03` to 16-bit binary.
2. Left circular shift by 6 bits. (Give Hex)
3. Right shift (RF) by 3 bits, then bitwise AND (&) with mask `0x0FFF`. (Give Binary)
4. Right circular shift by 14 bits. (Give Hex)
5. State which Left Circular Shift amount matches a Right Circular Shift of 14 bits and explain why.